

Angle Properties

WorksheetHub AI
Mathematics · Grade 7

Unit 3 · Geometry

Medium

Angles in lines, triangles and parallel lines

Name: _____

Date: _____

Score: / 12



Remember

Co-interior angles (C shape) between parallel lines add to 180° . The sum of the interior angles of a polygon with n sides is $(n - 2) \times 180^\circ$.

Straight line

angles add to 180°

Triangle

angles add to 180°

Around a point

angles add to 360°

Vertically opposite

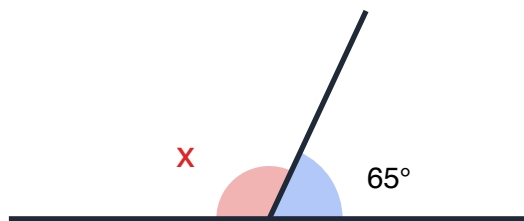
angles are equal (X)

Corresponding

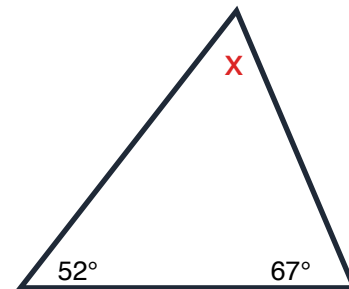
equal (F shape)

Alternate

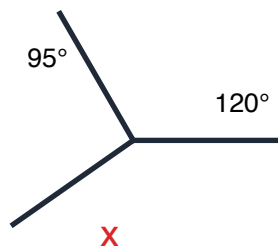
equal (Z shape)



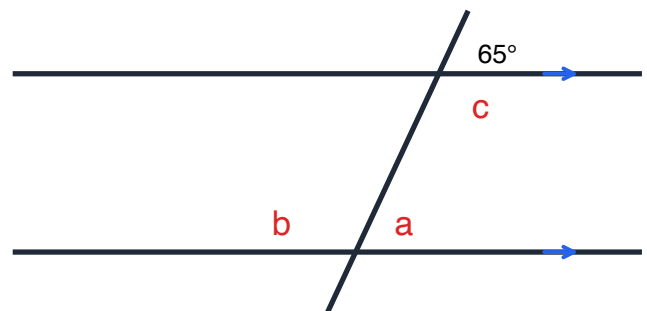
Angles on a straight line = 180°



Angles in a triangle = 180°



Angles around a point = 360°



Parallel lines (the arrows show they are parallel)

A. Find the missing angle x

- 1 Look at the straight-line picture: one angle is 65° . Find x .

Answer:

- 2 In the triangle picture, the other two angles are 52° and 67° . Find x .

Answer:

- 3 In the picture around a point, the angles are 120° , 95° and x . Find x .

Answer:

- 4 Two straight lines cross. One angle is 72° . Find the angle opposite it.

Answer:

B. Parallel lines (use the parallel-lines picture)

- 5 Find angle **a** and name the rule.

Answer:

- 6 Find angle **b** and name the rule.

Answer:

- 7 Find angle **c** and name the rule.

Answer:

C. Polygons

- 8 What is the sum of the interior angles of a pentagon (5 sides)?

Answer:

- 9 What is the sum of the interior angles of a hexagon (6 sides)?

Answer:

- 10 What is the size of each interior angle in a regular hexagon?

Answer:

★ Challenge

- 11 A triangle has angles x , $2x$ and $3x$. Find x and the size of each angle.

Answer:

- 12 Each exterior angle of a regular polygon is 45° . How many sides does it have?

Answer:

